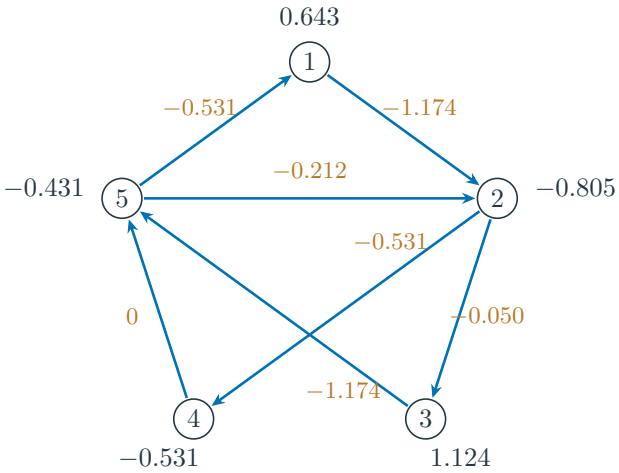


$$G_f(\theta)_x = \sum_{e=(p,x)} \ell_e - \sum_{e=(x,q)} \ell_e = (B\ell)_x$$

$$\begin{aligned} G_1 &= \ell_{5\to1} - \ell_{1\to2} \\ &= -0.531 + 1.174 = 0.643 \end{aligned}$$

$$\begin{aligned} G_2 &= \ell_{1\to2} + \ell_{5\to2} - \ell_{2\to3} - \ell_{2\to4} \\ &= -1.174 - 0.212 + 0.050 + 0.531 = -0.805 \end{aligned}$$

$$\begin{aligned} G_3 &= \ell_{2\to3} - \ell_{3\to5} \\ &= -0.050 + 1.174 = 1.124 \end{aligned}$$

$$\begin{aligned} G_4 &= \ell_{2\to4} - \ell_{4\to5} \\ &= -0.531 + 0 = -0.531 \end{aligned}$$

$$\begin{aligned} G_5 &= \ell_{3\to5} + \ell_{4\to5} - \ell_{5\to1} - \ell_{5\to2} \\ &= -1.174 + 0 + 0.531 + 0.212 = -0.431 \end{aligned}$$